

Cultural Homophily in Grand Slam Doubles

ATP Men's Grand Slam Doubles, 2018–2025 · 1,949 matches after dropping 48 retirements / walkovers · Regression sample: 3,752 team-obs (complete ranking, nationality/language for all 4 players; years since turning pro imputed to 0 for rookies)

1. Data and Measures

Three culture measures capture teammate similarity. **Same nationality** and **same official language** are binary (0/1). **Language proximity** is a continuous 0–1 index of ethnolinguistic closeness (higher = more similar language family).

Regressions use a team-level panel with tournament-by-year fixed effects, round fixed effects, and standard errors clustered by match. Controls in all main specifications: team average doubles ranking, opponent average ranking, top-100 singles indicator (either player), team average years since turning pro, demeaned (*exp_mean_dm*), and its square (*exp_mean_dm_sq*). Culture measures enter one at a time. Rank gap between teammates was tested as an additional control and is insignificant throughout; it is excluded from all main specifications. Nationality and language data are fully resolved in the pipeline (birthplace fallback, surname lookup, Monaco language correction, and `manual_nationality.csv`); zero matches are dropped for nationality.

2. Observation Breakdown

Step	Description	Matches	Change	Team-obs
1	Raw scraped dataset (2018–2025, incl. AO/RG/USO 2020; Wimbledon 2020 cancelled)	1,997	—	3,994
2	Drop retirements / walkovers (20 walkovers · 10 retired in set 1 · 18 retired in set 2+)	1,949	–48	3,898
3	Drop: doubles ranking incomplete for ≥1 player (4 of the original 14 were retrieved and restored — see note below)	1,939	–10	3,878
4	Drop: nationality / language missing for ≥1 player (pipeline fully resolves: birthplace, surname lookup, Monaco fix, <code>manual_nationality.csv</code>)	1,939	0	3,878
5	Exclude Olympic matches (Tokyo 2021 · Paris 2024)	1,876	–63	3,752
6	Years since turning pro: 213 obs are rookies (turned pro in/after the tournament year); <i>exp_mean</i> imputed to 0 — no obs dropped (these teams have mean rank ≈194 vs ≈122 for the rest; dropping would create selection bias)	—	0	3,752
→ Grand Slams regression sample (Tables 3–4)		—		3,752

Of the 14 matches originally dropped for incomplete doubles ranking, 4 were retrieved: 3 (`match_id` 6, 35, 49) failed only because Guillermo García-López's accented surname didn't match the unaccented scrape during the ranking merge — his doubles rank was matched in manually; 1 (`match_id` 913, 2021 Wimbledon) had a genuinely blank partner slot in the raw scrape (Alejandro Davidovich Fokina) — his full profile (rank, nationality, tenure, top-100 history) was reconstructed from public sources and the project's own gravity/ language lookup table. The remaining 10 could not be reliably retrieved and stay dropped.

Table 4 (tiebreak win): 1,197 tiebreaks × 2 = 2,394 team-obs (935 unique matches), covering sets 1–5 (main spec: the 1,182 standard 7pt tiebreaks; robustness spec adds 15 advantage-set/12-12-breaker deciders).

3. Pressure Outcome Counts

Computed on the **same 1,876-match regression sample** used in Tables 3–6 (not a broader pre-filter set), so every number below reconciles exactly with the regression tables. See §3.4 for how 1,876 relates to the raw data.

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	462	24.6%
Set-2 tiebreak (7-pt)	473	25.2%
Set-3 tiebreak (7-pt)	210	11.2%
Set-4 tiebreak (7-pt)	31	1.7%
Set-5 tiebreak (7-pt)	6	0.3%
Advantage-set / 12-12-breaker decider	15	0.8%
Any regular tiebreak (sets 1–2)	826	44.0%
Any tiebreak (all types, any set)	935	49.8%
Match went to 3 sets	829	44.2%
Comeback wins (winner lost set 1)	369	19.7%

Grand Slams regression sample, Olympics excluded (N=1,876) — §3.5 for the Olympics figures, reported separately since Olympic tennis follows different rules and isn't part of the regression sample. Sum of the five set-by-set standard-tiebreak rows: 462 + 473 + 210 + 31 + 6 = **1,182** — this matches Table 4's main-spec tiebreak count exactly, since both are now computed on the identical 1,876-match sample. **Any regular tiebreak (sets 1–2)** counts a match once if it had a tiebreak in set 1 and/or set 2 (a match with both counts once, not twice) — that's why it's less than 462+473. **Any tiebreak (all types, any set)** is the broadest category: a match counts once if it had a standard tiebreak in any of sets 1–5, or an advantage-set/12-12-breaker decider — it includes (doesn't add to) the "any regular tiebreak" matches, plus matches whose only tiebreak was in set 3/4/5 or was a decider. No tournament in this dataset ever records a genuine 10-point super-tiebreak as a set score — AO/RG/USO's set-3 score never exceeds 7 games. All 15 advantage-set/12-12-breaker cases are at Wimbledon (e.g. 16–14, 13–11, 12–10, 13–12), broken out precisely in §3.2 below.

3.1 Match Format: Best-of-3 vs. Best-of-5

Wimbledon played men's doubles as **best-of-5 sets in 2018, 2019, 2021, and 2022** (253 matches played under that rule within the regression sample, of which 101 actually required a 4th or 5th set — the rest finished in straight sets). Every other tournament-year in the Grand Slams — Australian Open, Roland Garros, and US Open in every year, and Wimbledon from 2023 onward — is best-of-3, with no match ever recording a 4th set. (Olympics is likewise best-of-3 throughout — see §3.5; Olympics is not part of the regression sample.)

Tournament	Years (best-of-5)	Matches (best-of-5)	Reached 4 sets	Reached 5 sets
Wimbledon	2018, 2019, 2021, 2022	253	101	50
All other tournament-years	—	Best-of-3 throughout	0	0

101, not 103: two matches previously counted as "reaching a 4th set" (match_id 504, 1204) are genuine mid-set-4 retirements, now correctly excluded from the sample — see §3.4.

3.2 Tiebreaks and Deciding-Set Outcomes by Set (1–5)

Three distinct outcomes end a set in this dataset. **Standard tiebreak** (ends 7–6/6–7, a 7-point breaker at 6-6) is the only outcome ever observed at AO/RG/USO, and applies to sets 1–2 everywhere plus any non-deciding set in a best-of-5 match. **12-12 breaker** (deciding set reaches 12 games apiece, then a breaker decides it, e.g. 13–12) and **advantage set**,

no breaker (deciding set won by a natural 2-game margin at 8+ games, e.g. 8–6, 11–9, 22–20) are observed only in deciding sets at Wimbledon (pre-2023).

Set	N reaching this set	Standard 7-6 TB	12-12 breaker	Advantage (no breaker)
1	1,876	462	0	0
2	1,876	473	0	0
3	829	210	2	8
4	101	31	0	0
5	50	6	1	7
Sum: 462 + 473 + 210 + 31 + 6	—	1,182	3	15

The 1,182 total exactly matches Table 4's main-spec tiebreak count, and the 1,876 matches reaching set 1 matches Section 2's regression sample size — both are computed on the identical sample. Set 3 now correctly shows both 12-12-breaker deciders (match_id 903, 927 — 2021 Wimbledon, both finished 13–12): a retirement-filter fix (§3.4) means the completeness check now recognizes a 1-game-margin breaker finish as valid, rather than misreading it as an incomplete/retired match.

3.3 Tiebreaks per Match

How many *standard* (7-pt) tiebreaks does a single match contain? A match can have more than one — e.g. a tiebreak in set 1 and another in set 2 — so this is a genuine distribution, not a binary yes/no. (The 15 advantage-set/12-12-breaker deciders are excluded here since they're a different category, and each can only ever be the very last set of a match, so they never combine with each other; a match can still have, e.g., a set-1 tiebreak *and* a 12-12-breaker decider in set 3.)

Tiebreaks in match	N matches	Share
0	948	50.5%
1	707	37.7%
2	190	10.1%
3	29	1.5%
4	2	0.1%
Total	1,876	100.0%

Check: $0 \times 948 + 1 \times 707 + 2 \times 190 + 3 \times 29 + 4 \times 2 = 1,182$ — matches the sum in §3.2 above exactly. Just over half of all matches (948, 50.5%) have no standard tiebreak at all; only 31 matches (1.7%) have three or more.

3.4 Where 1,876 and 1,886 Come From

All descriptive stats above (§3, 3.1–3.3) are computed on the exact 1,876-match regression sample, so they already reconcile with Tables 3–6 by construction. This section explains how that 1,876 relates to the raw data and the 48 dropped retirements/walkovers, and confirms a fix applied to the retirement filter.

Step	Matches
Raw scraped matches (Grand Slams + Olympics)	1,997
Drop 48 retirements/walkovers (47 GS + 1 Olympics)	1,949
↳ of which Olympic matches (Tokyo 2021 + Paris 2024)	63
↳ of which Grand Slam matches — this is where 1,886 comes from	1,886
Drop 10 more GS matches: incomplete doubles ranking (0 more for nationality/language) (4 of the original 14 were retrieved and restored — see §2)	−10
Final regression sample — this is 1,876, matching team_gs_panel.csv	1,876

1,886 = Grand Slam matches after dropping retirements/walkovers, but *before* the ranking/nationality-completeness filters — an intermediate count in the funnel, not used as the base for §3.1/3.2/3.3 above (those use 1,876). **1,876** = the final regression sample after also dropping the 10 remaining ranking-incomplete matches — this is the number §3.1/3.2/3.3 are built on.

Fix applied to the retirement filter (previously flagged as a known, unfixed gap): the completeness check only inspected sets 1–3, and only recognized a 2-game-margin finish (e.g. 8–6) as valid — not a 1-game-margin breaker finish (e.g. 13–12). Two Wimbledon 2021 matches (match_id 903, 927) that legitimately finished 13–12 in the third set were misread as incomplete and wrongly excluded; two other matches with a genuine retirement in the 4th set — Wimbledon 2019 (match_id 504, retired at 4–3) and Wimbledon 2022 (match_id 1204, retired at 1–1) — were invisible to a filter that never looked past set 3, and were wrongly included. Both issues are now fixed: the completeness check recognizes the 12-12-breaker pattern as valid, and sets 4–5 are now also checked when present. Net effect on the 1,876 count is a wash (2 matches swapped for 2 others), but the sample now contains the *correct* 2 in each direction. The advantage-set/12-12-breaker (match_tb) threshold was also widened from hi≥10 to hi≥8, since genuine long advantage-set finishes like 8–6 or 9–7 were being missed by the old threshold (7 such matches added: 4 at set 3, 3 at set 5).

Sample	Matches	Team-obs / tiebreaks
Pressure-outcome counts above (§3–3.3, GS regression sample)	1,876	—
team_gs_panel.csv (Table 3 sample, GS only)	1,876	3,752 team-obs
tiebreak_panel.csv (Table 4 sample, sets 1–5)	1,197 tiebreaks	2,394 team-obs

3.5 Olympics: Pressure Outcome Counts

Reported separately rather than pooled into §3 above, since Olympic tennis follows different rules in this dataset and Olympic matches are excluded from the regression sample entirely (Tables 3–6).

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	12	19.0%
Set-2 tiebreak (7-pt)	21	33.3%
Any regular tiebreak (sets 1–2)	28	44.4%
Advantage-set / 12-12-breaker decider (set 3)	20	31.7%
Any tiebreak (all types)	33	52.4%
Match went to 3 sets	20	31.7%
Comeback wins (winner lost set 1)	13	20.6%

N=63 matches (Tokyo 2021: 31, Paris 2024: 32). Best-of-3 throughout — no match reaches a 4th set. Unlike the Grand Slams, Olympic tennis has no genuine 7-6/6-7 set-3 finish at all in this dataset (0 of 20 three-set matches) — every three-set Olympic match is decided by the advantage-set/12-12-breaker rule (9 of 31 matches in 2021, 11 of 32 in 2024), never a standard tiebreak.

4. Olympic Cycle Descriptive Evidence

Table 2A — Tokyo 2021 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Tokyo	AO18 – USO19	1,004	40.0%	60.0%	54.8%	45.2%	55.7%	44.3%
Tokyo Prep	AO20 – Wim21 (ex Wim20)	680	40.3%	59.7%	55.0%	45.0%	55.1%	44.9%
Post-Tokyo	USO21 – USO22	602	39.0%	61.0%	54.8%	45.2%	55.6%	44.4%

N teams = GS matches × 2. Same/Diff = share of all teams (pooled winner + loser) on each cultural dimension.

Table 2B — Paris 2024 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Paris	AO22 – USO22	484	39.3%	60.7%	54.1%	45.9%	55.2%	44.8%
Paris Prep	AO23 – Wim24	860	42.6%	57.4%	57.6%	42.4%	58.0%	42.0%
Post-Paris	USO24 – USO25	626	43.9%	56.1%	58.8%	41.2%	60.2%	39.8%

N teams = GS matches × 2.

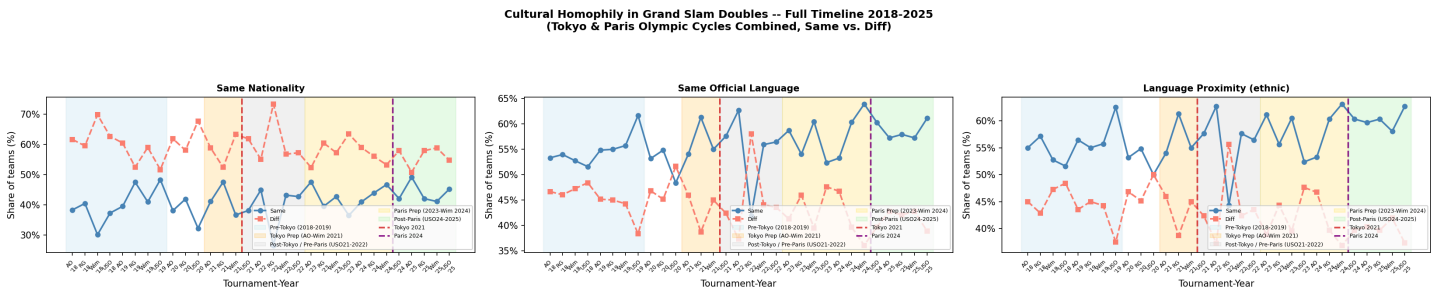


Figure 1. Both Olympic cycles on a single timeline (2018–2025), Grand Slams only. Vertical dashed lines mark Tokyo 2021 and Paris 2024.

5. Regression Results

Average marginal effects (AME, dP/dx) from logit models. Each column is a separate model. Standard errors in parentheses, clustered by match. FE: tournament × year, round. All specifications include years-since-turning-pro controls (exp_mean_dm , demeaned, and its square). Culture measures enter one at a time.

Table 3 — Outcome: Match Win | N = 3,752 team-obs | AME (dP/dx)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0371** (0.0163) p=0.023	+0.0469*** (0.0156) p=0.003	+0.0429*** (0.0156) p=0.006
Team avg. doubles rank	−0.0007*** (0.0001)	−0.0007*** (0.0001)	−0.0007*** (0.0001)
Opponent avg. doubles rank	+0.0007*** (0.0001)	+0.0007*** (0.0001)	+0.0007*** (0.0001)
Top-100 singles player	−0.0051 (0.0315) p=0.872	−0.0038 (0.0315) p=0.904	−0.0032 (0.0315) p=0.919
Team avg. yrs since turning pro	+0.0041** (0.0016) p=0.013	+0.0039** (0.0016) p=0.017	+0.0039** (0.0016) p=0.018
Yrs since turning pro, squared	−0.0007*** (0.0002) p=0.002	−0.0007*** (0.0002) p=0.002	−0.0007*** (0.0002) p=0.002
Pseudo-R ²	0.095	0.096	0.096

*** p<0.01 ** p<0.05 * p<0.10. Clustered SE. FE: tournament×year, round. All three culture measures statistically significant at the 5% level (nationality p=0.023, language p=0.003, proximity p=0.006). exp_mean is demeaned (exp_mean_dm = exp_mean − sample mean of 11.84 years since turning pro) and its square is built from the demeaned variable (exp_mean_dm_sq = exp_mean_dm²), applied consistently across every table in this report. This is a pure reparameterization — it leaves the fitted model, the culture AMEs, and every other control's AME unchanged; it only changes what "Team avg. yrs since turning pro" reads as (the marginal effect of tenure evaluated at the sample-mean tenure level rather than at zero tenure). Tenure shows a concave effect: positive linear (p<0.05) at the mean, negative quadratic (p<0.01). Top-100 singles indicator is not significant: within-GS doubles, having a top singles player does not predict match win once ranking and tenure are controlled.

Table 4 — Outcome: Tiebreak Win | Unit: team × tiebreak | AME (dP/dx)

Main specification: 7pt standard tiebreaks only, any set (1–5) — N = 2,364 team-obs (1,182 tiebreaks). **Robustness:** adds the 15 advantage-set/12-12-breaker match-tiebreak deciders — N = 2,394 team-obs (1,197 tiebreaks). Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq (same as Table 3; exp_mean demeaned against this panel's own estimation-sample mean: 11.99 main / 11.97 robustness). FE: tournament×year + round. SE clustered by match.

Variable	(1) Same Nationality		(2) Same Language		(3) Lang. Proximity	
	Main	Robust.	Main	Robust.	Main	Robust.
Culture measure	+0.0296	+0.0305	+0.0074	+0.0126	+0.0018	+0.0071
	(0.0223)	(0.0221)	(0.0218)	(0.0216)	(0.0217)	(0.0215)
	p=0.184	p=0.168	p=0.733	p=0.559	p=0.933	p=0.742
Team avg. doubles rank	-0.0003***	-0.0003***	-0.0003***	-0.0003***	-0.0003***	-0.0003***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles rank	+0.0003***	+0.0003***	+0.0003***	+0.0003***	+0.0003***	+0.0003***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player	-0.0661	-0.0595	-0.0633	-0.0568	-0.0631	-0.0566
	(0.0428)	(0.0425)	(0.0427)	(0.0424)	(0.0427)	(0.0424)
	p=0.123	p=0.161	p=0.139	p=0.181	p=0.140	p=0.182
Team avg. yrs since turning pro	+0.0025	+0.0025	+0.0022	+0.0022	+0.0021	+0.0021
	(0.0021)	(0.0021)	(0.0021)	(0.0021)	(0.0021)	(0.0021)
	p=0.235	p=0.243	p=0.302	p=0.303	p=0.314	p=0.315
Yrs since turning pro, squared	+0.0000	+0.0000	+0.0001	+0.0000	+0.0001	+0.0000
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
	p=0.870	p=0.922	p=0.816	p=0.871	p=0.806	p=0.859
Pseudo-R ²	0.014	0.013	0.013	0.013	0.013	0.013
N	2,364	2,394	2,364	2,394	2,364	2,394

Unit = one team per tiebreak (2 obs per tiebreak), now covering sets 1–5 (previously sets 1–3 only). Culture AMEs are small and insignificant across all three measures in both specs — homophily advantage does not extend to within-match tiebreak outcomes. Ranking controls remain strongly significant ($p < 0.001$), confirming the model captures real quality differences; tenure controls are not significant in the tiebreak sample. Main vs. robustness estimates are close throughout, so including the 15 advantage-set/12-12-breaker deciders does not change the conclusion.

6. Heterogeneity Analysis

Three sets of interactions are estimated on the main match-win sample ($N = 3,752$ team-obs), all with the full Table 3 control set (rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq): **Table 5**, culture $\times$ surface, testing surface-specific coordination demands; **Table 6**, culture $\times$ Hofstede individualism score (team-level, demeaned), testing whether collectivist teams gain more from cultural homophily; and **Table 6a**, culture $\times$ years since turning pro, testing whether experienced teams benefit more from cultural alignment.

Hofstede individualism scores (IDV) are taken from Geert Hofstede's official 6-dimension dataset (2015 release, geerthofstede.com). Countries not covered by Hofstede are assigned proxy scores following two tiers: *Tier 1* (direct cultural/linguistic kin, high confidence): BIH→SRB, CYP→GRC, MCO→FRA, MDA→ROU; *Tier 2* (same-region / closest available, moderate confidence): BLR→RUS, BOL→PER, DOM→COL, LBN→Arab composite (IDV=38), TUN→MAR, GEO→TUR, KAZ→RUS, UKR→RUS, UZB→RUS. The team-level score is $IC_team = (IDV_p1 + IDV_p2) / 2$, demeaned across the GS regression sample (mean = 62.98, SD = 20.85).

Table 5 — Heterogeneity: Culture × Surface | N = 3,752

Two specs, each pooling two surfaces into a single baseline rather than using all three at once. **Spec (a)**: grass vs. rest (hardcourt + clay pooled as baseline). **Spec (b)**: clay vs. rest (hardcourt + grass pooled as baseline). **No tournament×year FE** (round FE only) — with only two surface groups and no tournament×year FE, the surface main effect is directly estimable rather than absorbed, unlike the collinearity issue that arose with all three surfaces plus tournament×year FE together.

Variable	(1) Same Nationality		(2) Same Language		(3) Lang. Proximity	
	(a) grass	(b) clay	(a) grass	(b) clay	(a) grass	(b) clay
Culture (C) — pooled baseline	+0.0386** (0.0183) p=0.035	+0.0368* (0.0188) p=0.050	+0.0432** (0.0176) p=0.014	+0.0494*** (0.0181) p=0.006	+0.0409** (0.0176) p=0.020	+0.0418** (0.0180) p=0.020
Surface (grass in (a) / clay in (b))	+0.0062 (0.0148) p=0.675	−0.0007 (0.0142) p=0.963	−0.0063 (0.0206) p=0.760	+0.0060 (0.0188) p=0.748	−0.0024 (0.0207) p=0.909	−0.0022 (0.0194) p=0.909
C × Surface	−0.0080 (0.0373) p=0.831	−0.0003 (0.0346) p=0.993	+0.0142 (0.0361) p=0.694	−0.0108 (0.0340) p=0.750	+0.0078 (0.0362) p=0.830	+0.0035 (0.0344) p=0.918
Team avg. doubles rank	−0.0007*** (0.0001) p<0.001	−0.0007*** (0.0001) p<0.001	−0.0007*** (0.0001) p<0.001	−0.0007*** (0.0001) p<0.001	−0.0007*** (0.0001) p<0.001	−0.0007*** (0.0001) p<0.001
Opponent avg. doubles rank	+0.0007*** (0.0001) p<0.001	+0.0007*** (0.0001) p<0.001	+0.0007*** (0.0001) p<0.001	+0.0007*** (0.0001) p<0.001	+0.0007*** (0.0001) p<0.001	+0.0007*** (0.0001) p<0.001
Top-100 singles player	−0.0024 (0.0304) p=0.938	−0.0022 (0.0305) p=0.941	−0.0012 (0.0305) p=0.969	−0.0011 (0.0305) p=0.971	−0.0006 (0.0304) p=0.985	−0.0005 (0.0305) p=0.987
Team avg. yrs since turning pro	+0.0041** (0.0016) p=0.013	+0.0040** (0.0016) p=0.013	+0.0039** (0.0016) p=0.016	+0.0039** (0.0016) p=0.016	+0.0038** (0.0016) p=0.017	+0.0038** (0.0016) p=0.017
Yrs since turning pro, squared	−0.0006*** (0.0002) p=0.002	−0.0006*** (0.0002) p=0.002	−0.0006*** (0.0002) p=0.002	−0.0006*** (0.0002) p=0.002	−0.0006*** (0.0002) p=0.002	−0.0006*** (0.0002) p=0.002
Pseudo-R ²	0.095	0.095	0.096	0.096	0.096	0.096
N	3,752		3,752		3,752	

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq. FE: round only (no tournament×year FE). SE clustered by match. All surface main effects and interactions are insignificant (p > 0.69 throughout) in both specs. The culture advantage documented in Table 3 does not vary significantly by surface, consistent with the earlier three-surface specification.

Table 6 — Heterogeneity: Culture × Hofstede Individualism (demeaned) | N = 3,752

IC_team_dm = team-average Hofstede IDV score minus the sample mean (62.98). Positive values = more individualist than average. Hypothesis: more individualist teams rely less on in-group bonding → negative θ . Spec 2 only (includes the main IC effect) — the reported specification.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0301 [*]	+0.0397 ^{**}	+0.0356 ^{**}
	(0.0166) p=0.070	(0.0162) p=0.014	(0.0162) p=0.028
IC_team_dm	+0.0011 [*]	+0.0008	+0.0008
	(0.0006) p=0.055	(0.0008) p=0.298	(0.0008) p=0.291
C × IC_team_dm (θ)	−0.00049	−0.00002	−0.00000
	(0.00079) p=0.535	(0.00089) p=0.985	(0.00089) p=0.999
Team avg. doubles rank	−0.0007 ^{***}	−0.0007 ^{***}	−0.0007 ^{***}
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007 ^{***}	+0.0007 ^{***}	+0.0007 ^{***}
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0035	−0.0024	−0.0018
	(0.0316) p=0.911	(0.0317) p=0.940	(0.0317) p=0.954
Team avg. yrs since turning pro	+0.0048 ^{***}	+0.0046 ^{***}	+0.0046 ^{***}
	(0.0017) p=0.004	(0.0017) p=0.006	(0.0017) p=0.007
Yrs since turning pro, squared	−0.0006 ^{***}	−0.0006 ^{***}	−0.0006 ^{***}
	(0.0002) p=0.004	(0.0002) p=0.003	(0.0002) p=0.003
Pseudo-R ²	0.096	0.097	0.096
N	3,752	3,752	3,752

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq. FE: tournament×year + round. SE clustered by match. The culture main effects are consistent with Table 3. The interaction θ is small and insignificant across all three measures ($p > 0.53$), providing no evidence that individualism moderates the cultural homophily advantage. The main IC_team_dm coefficient is positive and marginally significant for same nationality ($p=0.055$) — teams with higher individualism scores perform somewhat better overall — but is insignificant for the language measures ($p>0.29$).

Table 6 robustness — dropping Tier-2 Hofstede-proxy countries | N = 3,597

Team-obs involving a player whose IDV score relies on a Tier-2 (moderate-confidence) proxy — BLR, BOL, DOM, GEO, KAZ, LBN, TUN, UKR, UZB — are dropped (155 of 3,752 team-obs), leaving only direct Hofstede matches and Tier-1 (high-confidence) proxies.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0212	+0.0312 [*]	+0.0271
	(0.0170) p=0.213	(0.0166) p=0.060	(0.0166) p=0.102
IC_team_dm	+0.0007	+0.0001	+0.0001
	(0.0006) p=0.253	(0.0008) p=0.950	(0.0008) p=0.946
C × IC_team_dm (θ)	+0.00008	+0.00084	+0.00086
	(0.00081) p=0.921	(0.00093) p=0.364	(0.00093) p=0.352
Team avg. doubles rank	−0.0007 ^{***}	−0.0007 ^{***}	−0.0007 ^{***}
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007 ^{***}	+0.0007 ^{***}	+0.0007 ^{***}
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0125	−0.0109	−0.0104
	(0.0329) p=0.703	(0.0330) p=0.742	(0.0330) p=0.753
Team avg. yrs since turning pro	+0.0049 ^{***}	+0.0047 ^{***}	+0.0047 ^{***}
	(0.0017) p=0.004	(0.0017) p=0.005	(0.0017) p=0.006
Yrs since turning pro, squared	−0.0006 ^{***}	−0.0006 ^{***}	−0.0006 ^{***}
	(0.0002) p=0.005	(0.0002) p=0.005	(0.0002) p=0.005
Pseudo-R ²	0.096	0.097	0.096
N	3,597	3,597	3,597

Same controls and FE as Table 6 above (exp_mean_dm carries the full-sample demeaning constant into this subsample rather than being re-centered), estimated on the Tier-2-excluded subsample. Point estimates on C shrink somewhat relative to the full-sample spec (e.g. same nationality: +0.0301 → +0.0212, no longer significant), consistent with the smaller, more homogeneous sample rather than a change in sign or a qualitatively different story. The IC_team_dm main effect for same nationality also loses significance (p=0.055 → p=0.253) once the weaker-confidence proxy countries are dropped, so that result should be treated as sensitive to the Tier-2 approximations. The θ interaction remains small and insignificant throughout, matching the full-sample conclusion that individualism does not moderate the cultural homophily advantage.

Table 6a — Heterogeneity: Culture × Years Since Turning Pro | N = 3,752

exp_mean is demeaned ($\text{exp_mean_dm} = \text{exp_mean} - \text{sample mean of 11.84 years since turning pro, SD} = 5.30$), consistent with how ic_team_dm is demeaned for Table 6. The main effect of culture (C) is therefore evaluated at the sample-average tenure level, not at zero; the interaction captures how the marginal effect changes per additional year since turning pro relative to that average. $\text{exp_mean_dm_sq} = \text{exp_mean_dm}^2$ (the demeaned quadratic control, used consistently across every table in this report) is included alongside exp_mean_dm, per the blanket rule that all heterogeneity tables carry the full Table 3/4 control set.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — mean experience	+0.0371 ^{**} (0.0163) p=0.023	+0.0445 ^{***} (0.0156) p=0.004	+0.0405 ^{***} (0.0156) p=0.009
C × exp_mean_dm	+0.0003 (0.0032) p=0.935	+0.0062 ^{**} (0.0031) p=0.048	+0.0061 [*] (0.0031) p=0.052
Team avg. doubles rank	−0.0007 ^{***} (0.0001) p<0.001	−0.0007 ^{***} (0.0001) p<0.001	−0.0007 ^{***} (0.0001) p<0.001
Opponent avg. doubles rank	+0.0007 ^{***} (0.0001) p<0.001	+0.0007 ^{***} (0.0001) p<0.001	+0.0007 ^{***} (0.0001) p<0.001
Top-100 singles player	−0.0052 (0.0315) p=0.869	−0.0067 (0.0314) p=0.831	−0.0060 (0.0314) p=0.847
Team avg. yrs since turning pro	+0.0040 [*] (0.0022) p=0.073	+0.0002 (0.0025) p=0.922	+0.0002 (0.0025) p=0.923
Yrs since turning pro, squared	−0.0007 ^{***} (0.0002) p=0.002	−0.0007 ^{***} (0.0002) p=0.002	−0.0007 ^{***} (0.0002) p=0.002
Pseudo-R ²	0.095	0.097	0.096
N	3,752	3,752	3,752

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq (full Table 3 control set). FE: tournament×year + round. SE clustered by match. With exp_mean_dm_sq back in the model, the culture main effects at mean tenure are close to Table 3's own estimates (+0.0371/+0.0469/+0.0429) and all significant. The positive and marginally significant interaction for language/proximity ($p \approx 0.048$ – 0.052) implies that advantage grows further with tenure above the mean; nationality homophily shows no significant tenure gradient ($p=0.935$).